

Climate Change Impact on India's Dairy Sector



Source: TH

Why in News?

A study published in **Scientific Reports** has found that **climate change -induced heat stress is significantly reducing milk production in the trans-Gangetic plains of northwestern India** , particularly in Haryana, one of India's major milk-producing regions.

What are the Key Findings of the Study Regarding Bovine Milk Production?

- **The Heat and Humidity Trap:** Temperatures exceeding **38°C** combined with relative humidity above **70%** (typical during the July-August monsoon overlap) trigger **catastrophic drops in milk yield**.
 - Winter temperature fluctuations showed negligible impact, establishing heat stress as the primary threat.
- **The Role of Potential Evapotranspiration (PET):** PET (a measure of atmospheric moisture demand) emerged as a critical indicator.
 - A single unit increase in PET slashes milk yield by roughly **1.4 liters per buffalo per day** .
- **Staggering Economic Baseline:** Heat-related stressors are already causing an annual national loss of **3.2 million tonnes of milk** , equating to roughly Rs 2,661 crore.
 - Without aggressive adaptation, this loss is projected to hit **15 million tonnes by the 2050s** .
- **The Lancet Projection:** Corroborating this data, *The Lancet* has projected that India's total milk production could plummet by up to **25% by 2085** due to unmitigated thermal stress.

Science of Heat Stress on Bovine

- When lactating animals cross their thermal comfort threshold, a high **Temperature-Humidity Index (THI)** triggers heat stress and disrupts normal physiological functions.
- Animals divert metabolic energy from milk production towards **thermoregulation through panting and sweating**, causing milk yields to decline by **20-30%**.
- Elevated body temperatures also increase **cortisol levels**, which impair the milk ejection reflex, while reduced feed intake creates a **negative energy balance** that further lowers productivity.
- Prolonged heat stress also weakens immunity, increasing livestock vulnerability to pathogens and parasites and raising farmers' veterinary expenses.

Differential Breed Vulnerability

- **Buffaloes (Extremely Vulnerable):** Dark skin and fewer sweat glands reduce heat dissipation, making them highly dependent on water wallowing for cooling.
- **Cross-Bred Cattle (Highly Vulnerable):** High metabolic heat production and poor adaptation to tropical heat make them susceptible to heat stress.
- **Indigenous Cattle (Highly Resilient):** **Bos indicus breeds** such as Sahiwal, Hariana and Gir possess loose skin, efficient sweating mechanisms and lower metabolic heat production, providing greater heat tolerance.

What are the Implications of Declining Milk Productivity on the Dairy Ecosystem?

- **Collapse of Rural Safety Net:** Dairy provides regular cash income to **80 million smallholder dairy farmers**, who contribute around **85% of India's milk production**.
 - Declining yields can increase indebtedness and livelihood insecurity, particularly among small and landless farmers dependent on mixed crop-livestock farming.
- **Spike in Food Inflation:** Milk and milk products have a significant weight in India's **Consumer Price Index (CPI)**. Supply contraction can raise retail milk prices and disproportionately affect low-income households.
- **Hit to Agricultural GVA:** The livestock sector contributes around **30% of Agricultural Gross Value Added (GVA)** making declining dairy productivity a threat to agricultural growth and farmers' incomes.
- **Threat to Dairy Cooperatives:** India's cooperative dairy model depends on aggregating small quantities of milk from millions of producers.
 - Falling per-animal yields can reduce procurement volumes and undermine the viability of village cooperatives and processing infrastructure.
- **Reversal of Women's Empowerment:** Women constitute **over 70% of India's dairy workforce**.
 - Declining dairy incomes can weaken women's financial independence, household decision-making power and progress towards **SDG 5 (Gender Equality)**.
- **Threat to Nutritional Security:** India's largely vegetarian population depends heavily on milk for

high-quality protein, calcium and Vitamin B12 .

- Reduced availability and higher prices can adversely affect children and pregnant women, undermining **POSHAN Abhiyaan** and **SDG 2 (Zero Hunger)** .
- **Disruption of the Nutrient Cycle:** Declining livestock populations can reduce the availability of **Farm Yard Manure (FYM)** , increasing dependence on chemical fertilisers and contributing to soil degradation, **n itrate pollution and loss of soil organic carbon**.
- **Vicious Climate Cycle:** Farmers may increase herd sizes to compensate for declining per-animal milk yields, increasing **enteric methane emissions** , **fodder demand and overgrazing** , thereby reinforcing climate change.
- **Crop Residue Mismanagement:** Livestock consume large quantities of straw, stalks and other agricultural residues.
 - A shrinking dairy ecosystem can reduce demand for such residues and potentially increase **stubble burning** and **air pollution in the Gangetic plains** .
- **Threat to Food Sovereignty:** India's average cattle **productivity (1,777 kg/animal/year) remains below the global average (2,699 kg), despite recording the world's highest productivity growth of 27.95% during 2013-20.**
 - Climate change could reduce India's milk production by 25% by 2085, threatening the self-sufficiency achieved through **Operation Flood** and increasing dairy imports, foreign exchange costs and food-security risks.

Initiatives to Strengthen the Livestock Sector

- **Rashtriya Gokul Mission**
- **National Livestock Mission (NLM)**
- **Animal Husbandry Infrastructure Development Fund (AHIDF)**
- **National Animal Disease Control Programme (NADCP)**
- **e-Pashuhaat & Pashu Sanjivni**

What Measures are Needed for a National Climate-Smart Dairy Strategy?

- **Climate Early-Warning Systems:** Integrate **Temperature-Humidity Index (THI) and Potential Evapotranspiration (PET)** into regional early-warning systems to predict heat stress and enable timely interventions.
- **Re-orienting Breeding Policies:** The Rashtriya Gokul Mission must pivot from purely yield-centric cross-breeding toward thermo-tolerant breeding programs.
 - Develop **thermo-tolerant breeding programmes** using heat-resilient traits of indigenous **Bos indicus breeds** , such as Sahiwal, Hariana and Gir.
 - Strengthen **in-situ conservation and farmer-led breed improvement programmes** , treating indigenous breeds as reservoirs of climate-resilient traits.
- **Climate-Smart Livestock Infrastructure:** Scale up infrastructure subsidies for smallholders under the

Animal Husbandry Infrastructure Development Fund (AHIDF) .

- Expand affordable access to **shaded sheds, wallowing ponds, sprinklers, foggers, misting systems and adequate drinking water** .
- Promoting agroforestry to create natural canopy shade for livestock, while simultaneously cultivating drought-resistant forage crops to secure feed supply against erratic monsoons.
- **Risk Financing and Insurance:** Develop **heat-index-based livestock insurance** , affordable credit and targeted subsidies to protect smallholder farmers against climate-induced productivity losses.

Conclusion

Climate change poses a systemic threat to India's dairy sector, rural livelihoods and food sovereignty. Protecting the legacy of the **White Revolution requires a "White Revolution 2.0" centred on climate resilience** , integrating indigenous heat-tolerant breeds, climate-smart infrastructure and meteorological early-warning systems. Such a transition is essential for securing rural incomes, gender equity and nutritional security through a **One Health approach** .

Drishti Mains Question:

Climate change is transforming heat stress from a livestock health issue into a systemic threat to India's rural economy and food security. Discuss.

Frequently Asked Questions (FAQs)

1. What climatic conditions significantly reduce bovine milk production?

Temperatures above **38°C combined with relative humidity exceeding 70%** trigger severe heat stress, reducing feed intake, disrupting milk ejection and lowering milk productivity.

2. Why are indigenous cattle breeds considered important for climate-resilient dairy farming?

Bos indicus breeds , such as Sahiwal, Hariana and Gir, possess efficient sweating mechanisms, lower metabolic heat production and other thermo-tolerant traits that provide greater resilience to heat stress.

3. What is the role of THI and PET in climate-smart livestock management?

Temperature-Humidity Index (THI) measures livestock heat stress, while **Potential Evapotranspiration (PET)** reflects atmospheric moisture demand. Their integration into early-warning systems can enable proactive livestock management.

4. Which government initiatives support the development of India's livestock and dairy sector?

Major initiatives include the **Rashtriya Gokul Mission, National Livestock Mission (NLM), Animal Husbandry Infrastructure Development Fund (AHIDF), National Animal Disease Control Programme (NADCP), e-Pashuhaat and Pashu Sanjivni** .

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UPSC Civil Services Examination, Previous Year Questions (PYQs)

Prelims

Q. As per the NSSO 70th Round “Situation Assessment Survey of Agricultural Households”, consider the following statements: (2018)

1. Rajasthan has the highest percentage share of agricultural households among its rural households.
2. Out of the total agricultural households in the country, a little over 60 percent belong to OBCs.
3. In Kerala, a little over 60 percent of agricultural households reported to have received maximum income from sources other than agricultural activities.

Which of the statements given above is/are correct?

- (a) 2 and 3 only
- (b) 2 only
- (c) 1 and 3 only
- (d) 1, 2 and 3

Ans: (c)

Mains

Q. Livestock rearing has a big potential for providing non-farm employment and income in rural areas. Discuss suggesting suitable measures to promote this sector in India. **(2015)**

World Bank Income Classification: India Remains Lower-Middle-Income



Source: BT

Why in News?

World Bank Group Country Income Classifications for Fiscal Year 2027 (released in July 2026) have maintained India's status as a '**Lower-Middle-Income**' economy, a status it has held since **2009**.

- In a contrasting regional development, **Sri Lanka** (along with Vietnam and the Philippines) was

upgraded to the **Upper-Middle-Income** category this year, successfully bouncing back from its severe 2022 economic crisis.

What are the Key Highlights of the World Bank Country Income Classification for Fiscal Year 2027?

- **Upgraded Economies:** Six nations crossed into higher income categories this fiscal year. In **1987**, **nearly 30% of economies were classified as low-income**, which declined sharply to **11% by 2026**, reflecting a significant shift in the global income landscape.
 - **Lower-Middle to Upper-Middle Income:** Sri Lanka, Vietnam, the Philippines, Jordan, and Micronesia.
 - **Low to Lower-Middle Income:** Togo.
- **Diverse Growth Trajectories:**
 - **Vietnam & Philippines:** Driven by export-led industrial growth and sustained expansions in **Gross National Income (GNI)**.
 - **Sri Lanka:** Demonstrated notable post-crisis resilience, rebounding via tourism and financial service revivals after its recent sovereign debt default.
 - **Jordan & Togo:** Primarily aided by statistical adjustments, including updated national accounts methodologies and revised census data.
- **India's Position:** India remains steady in the **Lower-Middle-Income** bracket, with a national income (GNI) per capita at USD 2760 in 2025.

How Does the World Bank Classify Economies?

- **Country Income Classifications:** The World Bank classifies the world's economies into four income groups: **Low, Lower-Middle, Upper-Middle, and High-Income**.
 - The World Bank classifies the world's economies annually on **1 st July**. The classification is based on the **Gross National Income (GNI) per capita** of the previous calendar year.
- **The Atlas Method:** To ensure accuracy and cross-country comparability, the GNI is expressed in current US dollars using the **Atlas method**.
 - This method is a special exchange rate smoothing technique designed to reduce the impact of exchange rate fluctuations and inflation in cross-country comparisons.
 - A country must experience real economic growth that outpaces population growth and inflation to move up a category.
- **Annual Updates:** The threshold for each category is adjusted annually in July using the **Special Drawing Rights (SDR) deflator to account for global inflation**.
- **Current Brackets for 2026-2027 (FY27):**

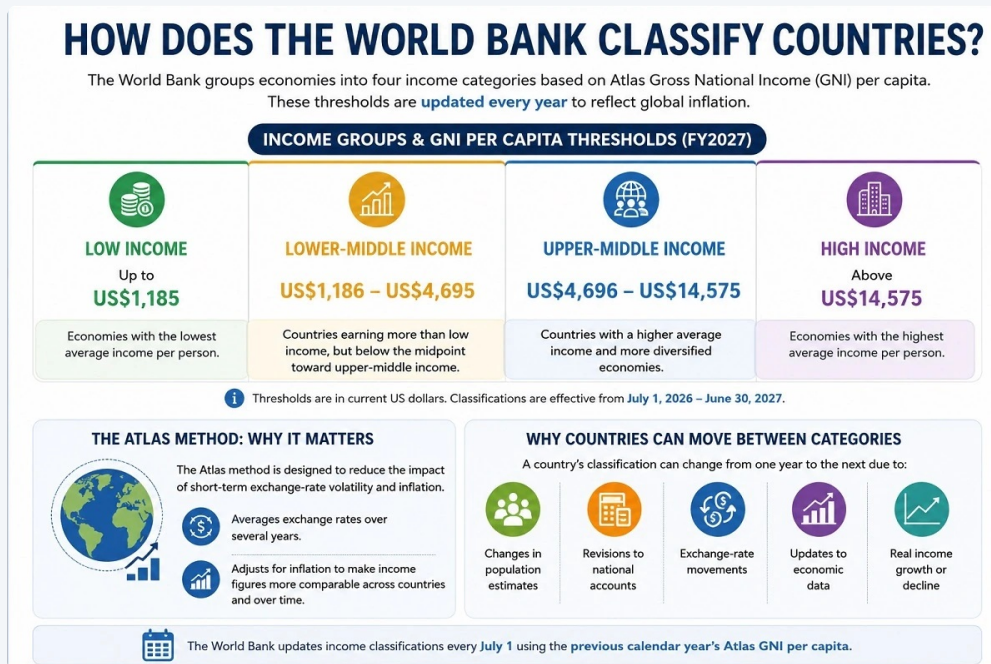
Income Category	GNI Per Capita (2025)
Low-Income	USD 1,175 or less
Lower-Middle-Income	USD 1,176–USD4,635
Upper-Middle-Income	USD 4,636–USD14,375
High-Income	More than USD 14,375

- **Significance:**

- **Access to World Bank Loans:** Low-income countries can receive concessional loans and grants from [International Development Association \(IDA\)](#) , while many middle-income countries borrow from [International Bank for Reconstruction and Development \(IBRD\)](#) on less concessional terms.
- **Allocation of Foreign Aid:** Donor countries and international organizations use income levels to determine **which countries should receive development assistance** .
- **International Trade Benefits:** Income status may influence eligibility for **preferential trade schemes and special treatment for developing countries** .
- **Distribution of Climate Finance:** Helps international institutions prioritize **financial support for economically vulnerable countries** to address climate change.
- **Investor Assessment:** Income classification provides a broad indicator of a country's **economic development level** , which investors may consider alongside other factors.
- **Policy Formulation:** Governments use income categories to **compare economic progress with other countries** and formulate policies to avoid the **Middle-Income Trap** .

Limitations of GNI per Capita

- GNI per capita is a **crude average that does not measure overall development**, as it masks income inequality, wealth concentration and the depth of poverty while ignoring health, education, institutional strength, environmental sustainability and unpaid or informal work.
- The **World Bank's income classification is primarily an analytical and operational tool, not a measure of a country's overall development**.



Why Does India Continue to Retain 'Lower-Middle-Income' Status?

- **The Denominator Challenge:** India's immense population base of over 1.4 billion acts as a massive denominator.
 - Even robust aggregate GDP expansions get diluted when converted to per capita earnings, making the leap to the upper-middle threshold mathematically steep.
- **Stark Regional and Intra-State Disparities:** Aggregate figures conceal massive domestic economic divergence.
 - Industrialized, high-productivity states (such as Tamil Nadu, Maharashtra, and Gujarat) exhibit **per capita incomes far outstripping the national average**. Conversely, highly populous states (like Bihar and Uttar Pradesh) experience **low productivity and income growth**, dragging down the national statistical mean.
- **Informalization and Disguised Unemployment:** Up to **85% of India's workforce remains in the informal sector**, with a heavy, low-productivity reliance on agriculture.
 - This suppresses average earnings, limits the growth of aggregate wage income, and hinders the expansion of the formal tax base.
- **Historical Starting Line:** India transitioned out of the "Low-Income" bracket relatively recently in **2009**

- Crossing into the next classification tier inherently requires doubling or tripling per capita income from its historical baseline - a structural evolution that historically requires decades of unbroken, high-percent real growth.

Frequently Asked Questions (FAQs)

1. How does the World Bank classify countries into different income groups?

The World Bank annually classifies economies into **Low, Lower-Middle, Upper-Middle and High-Income groups** based on the previous calendar year's **GNI per capita calculated using the Atlas Method** .

2. What is the World Bank Atlas Method?

The **Atlas Method** converts GNI into US dollars using a three-year average exchange rate adjusted for inflation, reducing the impact of short-term exchange-rate fluctuations on cross-country comparisons.

3. Why does India remain a Lower-Middle-Income Economy?

India's transition is constrained by its **large population denominator, regional income disparities, low labour productivity, high informal employment and dependence on low-productivity agriculture** .

4. What is the significance of the World Bank's income classification?

Income classifications influence access to **IDA and IBRD financing, development assistance, concessional finance, climate finance and international economic comparisons** .

[Watch Video on YouTube:

▶ <https://www.youtube.com/embed/-izCjFX-clQ>

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UPSC Civil Services Examination, Previous Year Questions (PYQs)

Prelims

Q. India's ranking in the 'Ease of Doing Business Index' is sometimes seen in the news. Which of the following has declared that ranking? (2016)

- (a) Organization for Economic Cooperation and Development (OECD)
- (b) World Economic Forum
- (c) World Bank
- (d) World Trade Organization (WTO)

Ans: (c)

Q. Increase in absolute and per capita real GNP do not connote a higher level of economic development, if: (2018)

- (a) Industrial output fails to keep pace with agricultural output.
- (b) Agricultural output fails to keep pace with industrial output.
- (c) Poverty and unemployment increase.
- (d) Imports grow faster than exports.

Ans: (c)

Q. In a given year in India, official poverty lines are higher in some States than in others because: (2019)

- (a) Poverty rates vary from State to State
- (b) Price levels vary from State to State
- (c) Gross State Product varies from State to State
- (d) Quality of public distribution varies from State to State

Ans: (b)

Mains

Q.1 "Industrial growth rate has lagged behind in the overall growth of Gross-Domestic-Product(GDP) in the post-reform period" Give reasons. How far the recent changes in Industrial Policy capable of increasing the industrial growth rate? **(2017)**

Q.2 Normally countries shift from agriculture to industry and then later to services, but India shifted directly from agriculture to services. What are the reasons for the huge growth of services vis-a-vis the industry in the country? Can India become a developed country without a strong industrial base? **(2014)**